Developing Machine Learning Models to Detect Breast Cancer

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***Abstract*—Breast cancer is the most common cancer in females, causing a significant amount of morbidity and mortality. In 2020, there were about 2.3 million new cases and 684,996 deaths worldwide. Early detection is critical, especially in younger women in India, where incidence rates are rising. This study explores the use of artificial intelligence and machine learning for early-stage breast cancer detection and survival prediction using the Breast Cancer Wisconsin dataset. It points out key characteristics of cell nucleus based on the feature extractions from FNA images of breast masses. Various distinct techniques of machine learning have been addressed in this study, including Random Forest, K-Nearest Neighbors, Support Vector Machines, and Logistic Regression. Out of these, Support Vector Machines and Logistic Regression produced similar results, with accuracy values of 98.24%. These results demonstrate the possibility of enhanced outcome measures and promotion of early diagnosis for the patients through machine learning.**

Keywords—Breast Cancer, Machine Learning Models, Logistic Regression, k-nearest neighbor algorithm (KNN), Random Forest, Support vector machine (SVM).

I. Introduction

As per the report by CDC (Centers for Disease Control and Prevention), breast cancer is one of the cancers. Except for buccal cavity malignancy, which has higher deadliness. Cancer is the most problematic issue across the globe; both external and internal contribute to neoplastic development, including but not limited to metabolic, genetic, transcriptomic etc., [1], [2], [3], [4] In the year 2020, there were an estimated 2.3 million new breast cancer cases and 684,996 breast cancer deaths among women worldwide. Affecting females at the ratio of 1-in-8 while very few men are affected. For diagnosis/treatment, special procedures like mastectomy, chemotherapy, radiotherapy, and hormone therapy [2], [5]. Breast cancer is defined as a type of cancer that starts in the tissue of the breast and causes excessive growth of breast tissues, resulting in tumors. Uncontrolled cell proliferation causes malignant tumor formation, which invades and damage surrounding healthy tissue. The positive lymph nodes suggest their involvement and indicate cancer spread.

For breast cancer survival, factors such as environmental chemicals, reproductive factors, lifestyle behaviours, drinking alcohol, sedentary lifestyle, and obesity play a major role [6], [7]. Also, some gene mutations that are found in the germline are associated with an increased incidence of breast cancer owing to genetic loss, DNA methylation, and gene mutations [8]. And this is where the power of Artificial Intelligence and Machine learning comes in because to reach such a conclusion and make a decision on the presence of the disease requires multi-faceted interrelationship of factors. Cognitive capabilities of AI and machine learning systems are perhaps the most sought after in today's world due to their ability to replicate human intelligence, in that they can evolve with the addition of knowledge over time. These accessories contain power algorithms that help to analyze the accessible data and therefore the projection can be done more specifically despite the nature of the data being images, tables, and so forth. Exploring these possibilities enables opportunities to rethink and innovate new avenues for treatment by employing AI and ML for biological development. Subjects were fed mammogram images in a study, which has previously confirmed that Older Deep Learning-based Architectures are beneficial for Breast cancer diagnosis [9]. These techniques are logically substantiated and can be justified in biomedical contexts. Before we continue with the discussion on the engineering applications, we will first address the molecular biology of the most important prognostic factors in this pathology. The erratic growth of cells in the breast is referred to as breast cancer. Even though breast cancer has different forms, its type will depend on the cell in the breast that has turned cancerous. The cancer-affected breast is composed of three broad areas: the fibrous, the ductal, and the lobular tissues, whereby the cancer can begin in any of these areas. However, most of breast cancers arise from either the lobules or the ducts and they invade the adjoining blood and lymphatic vessels which is known as metastatic spread. The common

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types of breast cancer are Invasive ductal carcinoma, which starts from the ducts, and Invasive lobular carcinoma, relatively where the cancers start from the lobules. Early detection of breast cancer is often treatable, as most cases go unnoticed until the disease is advanced, chronic, and life-threatening. Artificial Intelligence has been another major factor and machine learning is a technology which enables the computer in achieving intelligence like that of a human being and also improving its level of operation. The medical field is one of the sensitive domains where critical decisions with accuracy must be made to cure diseases. Minor mistakes in making the correct decision may affect the lives of patients. Artificial Intelligence and Machine Learning can effectively detect and identify the probable breast cancer. Early diagnosis followed by proper treatment. leads to curing the disease and increases the possibility of patients' survival. Despite existing methodologies for detection, the importance of this framework can be justified as it can improve the outcome, which could be enhanced by further categorization if the genetic information is tailored accordingly. The data used for the experiment was acquired from open-source platforms, and we opted for the "Breast Cancer Wisconsin (Diagnostic) Data Set" [10] hereafter referred as dataset, which is well known medical dataset for binary classification tasks, we worked on Logistic Regression (LR), K-nearest neighbor (KNN), Support Vector Machine (SVM), Random Forest (RF) for detecting the tumor and the disease. Dataset contains data on 569 instances of breast cancer cases, where each instance is classified as either maligned which means cancerous or benign which means non-cancerous. The dataset includes 30 real-valued input features derived from a digitized image of a breast mass fine needle aspirate (FNA). These features are computed from the images and describe the characteristics of cell nuclei, such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension. Each feature is averaged with standard error and worst value calculated. The target variable which represents the diagnosis encoded as 1 for malignant and 0 for benign. A better execution and improved strategies prove to be more valuable for patients and increase the likelihood of medical success in artificial intelligence domain

II. RESULTS

This section describes the analysis of breast cancer data using machine learning models.

We chose LR, KNN, RF, and SVM models because they are appropriate for the dataset, containing crucial biomarkers such as radius, texture, and concavity. LR is a suitable model for binary classification and interpretable, thus it helps correlate clinical features with biological markers [11]. KNN is a non- parametric algorithm that finds patterns in non-linear data, making it very useful for analyzing complex biological variations [12]. RF, with its ensemble learning, handles overfitting and captures intricate feature relationships [13]. SVM creates precise decision boundaries, performing well in high- dimensional datasets, thereby ensuring robust diagnosis predictions [14]. Preprocessing steps included normalization of feature values and handling missing data to improve the accuracy and efficiency of the model. These algorithms were selected to balance interpretability, computational performance, and their ability to integrate computational findings with clinical implications.

*A. **Logistic Regression** (LR):*

A LR model was applied to the dataset to predict whether a tumour is benign or malignant based on all 30 input features. For this particular dataset, the LR statistical model used for classification tasks is a more practical choice. In this algorithm, the decision boundary (linear) is first defined in the feature space and a logistic sigmoid function (σ) is applied to obtain a value between 0 and 1 [Fig.1].

$$\sigma(x) = 1 / 1+e^{(-x)} \quad (1)$$

Thus, the model predicts the chances of any occurrence being classified as benign (0) or malignant (1). To build the model and evaluate the model performance on a different dataset, the data was split into training and testing sets. The model effectiveness was evaluated using performance measures like recall, accuracy, precision, and F1 score [TABLE 1]. Due to the well-structured data, logistic regression which is the relationship of features and the log odds of the output was able to segregate data into benign and malignant tumors.

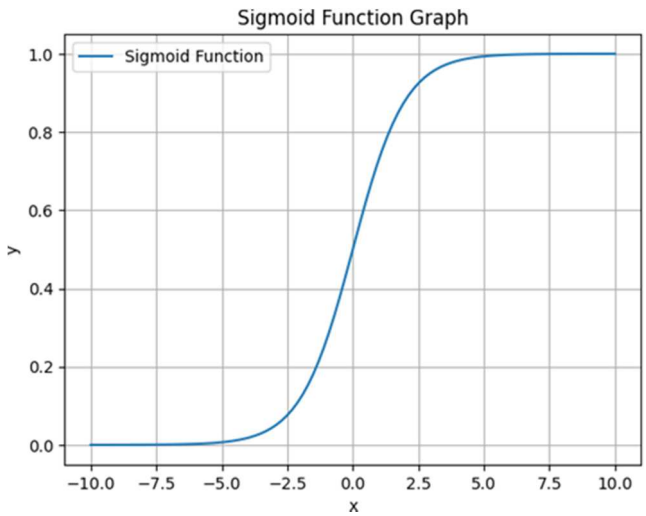

Fig. 1. The graph of sigmoid function

TABLE I. LOGISTIC REGRESSION RESULTS

	precision	*recall*	*f1-score*	*support*
0	0.97	0.95	0.98	108
1	0.98	0.99	0.97	63

*B. **K-nearest neighbor*** (KNN)

Using dataset, a KNN model was developed in order to classify whether a tumor is benign or malignant. KNN is a type of instance based learning where the test instance is compared to the nearest 'k' instances in the d-dimensional feature space of the training data denoted by the user. Then, for a test

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instance the label malignant or benign is assigned class which is in the majority of the k nearest neighbors [15]. The relationship between the error rate and different K-values in KNN model fluctuates as K increases, reaching its minimum around K=6 before stabilizing [Fig.2].

$$d(Xi, Xj) = \sqrt{\sum_{k=1}^{n} (X_{ik} - X_{jk})^2} \qquad (2)$$

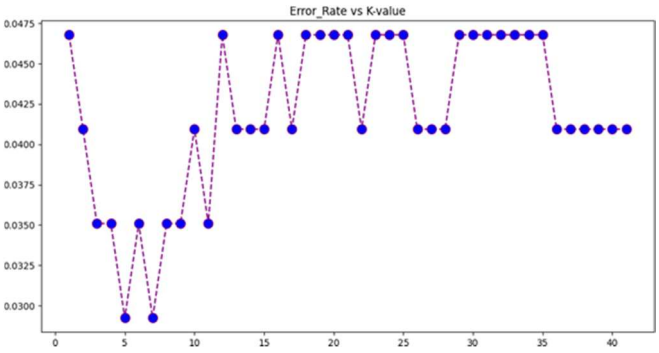

Fig. 2. Show the mean error for different K values

In order to achieve the best performance, cross-validation/holdout approaches was done to find the best k. "k" being one of the most important hyper-parameter that affects the performance of the model. Too high "k" might lead to under-fitted models while too low k's can lead over-fitted s model which are prone to noise. In Additions the dataset was randomized and divided in to training and test sets in order to facilitate testing without using the already seen data. KNN algorithm makes no assumption about the distribution of the underlying data which suits worst cases of simple or even complex parameters' non-linearity. The downside of the model is that it can be time-consuming for large databases since the distances from each point to all the other points in the data base have to be computed. Other evaluation metrics such as accuracy, precision, recall and F1 score were computed [TABLE.2] in order to evaluate the KNN modeling performance.

TABLE II. K NEAREST NEIGHBOR RESULTS

	precision	**recall**	**f1-score**	**support**
0	0.96	0.98	0.97	108
1	0.97	0.94	0.95	63

C. Random forest Model (RF)

In addition to the aforementioned approaches, RF model was fitted to the dataset, in order to increase the chances of predicting whether the tumors are malignant or benign. In contrast to LR which applies whenever variables are linearly dependent and KNN which uses proximity to make the decision, RF is a very effective ensemble model that combines many decision trees. This is achieved by training every tree on a unique random sample of the data and features, therefore combating overfitting, which enhances the ability of the model to learn new information. Just as how LR is easily understood using its feature weights and KNN understand the closeness of the given data points, RF understands and learnt intricate interactions which may exist in the data. In order to achieve equal comparison among the models, they were all subjected to the same training and testing data splits. The evaluation was based on important indicators, such as Accuracy, precision, recall, and the F1 score for RF models [TABLE.III].

TABLE III. RANDOM FOREST RESULTS

	precision	***recall***	***f1-score***	***support***
0	0.96	0.99	0.98	108
1	0.98	0.94	0.96	63

Among the many merits of the RF is the feature importance measure, which helps to pinpoint those tumor cell attributes that are most predictive of cancer. The normalized confusion matrix [Fig.3] illustrates the performance of the classification model, with a high accuracy observed for Class 1 (99%) and Class 2 (93%) predictions, showcasing the model's effectiveness in distinguishing between the two classes. Hyper-parameters like the number of trees and the maximum depth of each tree were varied in order to enhance the performance of the model.

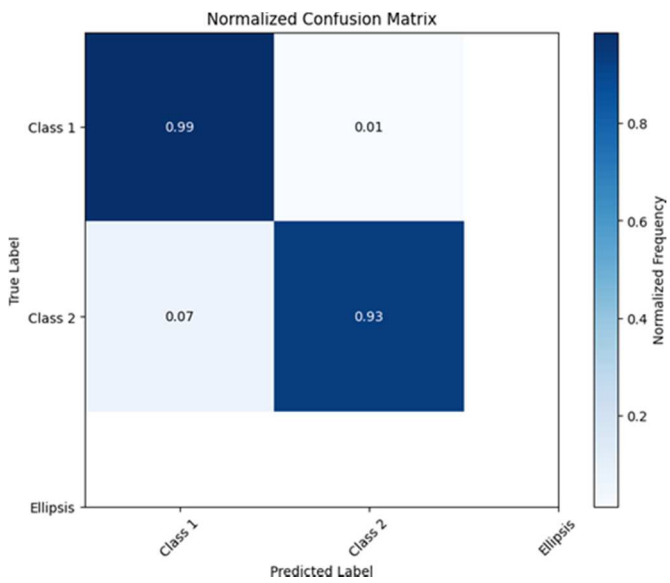

Fig. 3. Show Confusion matrix for a random forest classifier, showing the distribution of true and predicted labels

D. Support Vector Machine (SVM)

The dataset implemented SVM in order to classify data as benign or malignant tumors. SVM are sophisticated supervised learning algorithms that perform classifications while attempting to construct an optimal hyperplane which

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separates classes with the maximum distance from any data point to the hyperplane of the opposite class. This method is rather effective for this sort of data, especially when the data is 30-dimensional as in this case, which is very useful. SVM, similar to the RF algorithm, helps identify patterns and structures that may not be visible in the data. However, unlike LR, SVM is not constrained on the space with a straight line or a curve because of its kernel functions like Radial Basis Function (RBF). Therefore, it can work with multiple dimensions that are not necessarily linear. In order to ensure the uniformity in the assessment process, SVM model evaluation as well as tuning was performed with the same data partitions that were used in simulation of other models for this study. The success of SVM was measured with the parameters of accuracy, precision, recall and F1[TABLE.IV], which made it possible to compare it directly with RF, KNN and LR models.

TABLE IV. SUPPORT VECTOR MACHINE RESULTS

	precision	***recall***	***f1-score***	***support***
0	0.98	0.99	0.99	108
1	0.98	0.97	0.98	63

In order to improve the performance of the model other also worked including the choice of kernel and regularization parameter C. The ability of SVM to yield different decision regions, as well as its applicability to linear and non-linear relations, presents a positive aspect in the collection of models used for breast cancer diagnosis.

III. CONCLUSION

This study applied several machine learning algorithms, namely LR, KNN, RF, and SVM, to classify the tumors as malignant or benign in the dataset. Of these algorithms, SVM had the highest accuracy in identifying the patterns of intricate, nonlinear relationships in the data at 98.24% [TABLE.V]. Another technique, Logistic Regression, was found to be slightly less accurate (97.66%) but was easy to interpret and thus valuable for clinical decision-making applications. While KNN was as simple and easy to implement, it was able to still achieve a 96.49% accuracy score. Random Forest (97.07%) provided insight into how features are important, showing that such characteristics of tumor as radius and texture can be strong indicators.

TABLE V. ACCURACY SCORE RESULTS OF ALL THE MODELS UNDER STUDY

Algorithm	**Accuracy score**
Logistic Regression	97.66%
K-nearest neighbor	96.49%
Random Forest Model	97.07%
Support Vector Machine	98.24%

Therefore, Machine Learning models can be a futuristic approach to saving time and improving the accuracy of disease prediction. Health science is a vast and data-driven field, and the usage of these AI/ML models can lead to satisfying results. Advanced methods, like incorporating advanced deep learning architectures or feature engineering techniques, can be the next step for future work, like gaining more performance improvement in prediction and bringing such models into a broader area of clinical workflow.

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